BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021 • Paper Code: DIK-HSSC-XI-2021(A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 18 | **Paper Code:** DIK-HSSC-XI-2021(A)

Q1. Choose the correct option i.e. A,B,C,D and fill it up in the correct bubble in the given MCQs sheet.

Qi. At terminal velocity the acceleration of falling body is _____

(A) $0ms^{-2}$
(B) $4.9ms^{-2}$
(C) $-9.8ms^{-2}$
(D) $9.8ms^{-2}$

Qii. Substance that does not flow easily (like honey) have the _____ value for co-efficient of viscosity.

(A) Low
(B) High
(C) Zero
(D) Negative

Qiii. Tuning of radio set is an example of _____

(A) Mechanical resonance
(B) Musical resonance
(C) Electrical resonance
(D) Free vibrations

Qiv. Deriving the equation of time period for simple pendulum which quantity should be kept small.

(A) Length of simple pendulum
(B) Amplitude
(C) Mass of simple pendulum
(D) Gravitational acceleration

Qv. Wave generator produces 500 pulses in 10 sec. Find the frequency of the pulses produce in second [illegible in source scan]

(A) 5000
(B) 500
(C) 50
(D) 1/50

Qvi. Which of the following factor has no effect on the speed of sound in gas?

(A) Humidity
(B) Pressure
(C) Temperature
(D) Density

Qvii. Coloured fringes observed in a soap bubbles are examples of _____

(A) Diffraction
(B) Interference
(C) Polarization
(D) Reflection

Qviii. The S.I unit of molar specific heat or molar specific capacity is given by _____

(A) J $mole^{-1}$
(B) J/Kg
(C) $Jmole^{-1}k^{-1}$
(D) $Kg.molek^{-1}$

Qix. The dimension of power is _____

(A) $[ML^2T^3]$
(B) $[ML^2T^{-3}]$
(C) $[MLT^3]$
(D) $[MLT^{-3}]$

Qx. For which angle the equation |A.B| = |AxB| is correct?

(A) 30°
(B) 45°
(C) 60°
(D) 90°

Qxi. To improve the jumping record the long jumper should jump at the angle of _____

(A) 30°
(B) 45°
(C) 60°
(D) 90°

Qxii. Range of projective on a horizontal plane is same for the following pair of angles.

(A) 15° and 18°
(B) 43° and 47°
(C) 20° and 80°
(D) 52° and 62°

Qxiii. If Momentum is increased by 50% then KE increases by:

(A) 25%
(B) 100%
(C) 125%
(D) 200%

Qxiv. If Ax = 1.5cm, Ay = −1.0cm into which quadrant do the vector A points?

(A) I
(B) II
(C) III
(D) IV

Qxv. Value of the escape velocity is given as _____

(A) 11.2m/s
(B) 11.2 x 10^2 m/s
(C) 11.2 x 10^3m/s
(D) 112 m/s

Qxvi. The atmosphere is held to the earth by_____

(A) Wind
(B) Gravity
(C) Cloud
(D) Rotation of earth

Qxvii. Moment of inertia of thin walled cylinder or ring is given by:

(A) MR^2
(B) $1/2\ MR^2$
(C) $1/4\ MR^2$
(D) $22/7\ MR^2$

Qxviii. The angular speed in radians/hour for daily rotation of our earth is_____

(A) 2π
(B) 4π
(C) $\pi/6$
(D) $\pi/12$

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 67

Q2. (Marks 40) Answer any TEN questions of the following. All questions carry equal marks: [40]

i. Show that L = Lw.

ii. Describe Brewster's law of polarization.

iii. Calculate the angle for which the maximum height reached and corresponding range are equal.

iv. Explain the phenomenon of reflection of sound waves.

v. State and explain the first law of thermodynamics.

vi. What is Stocke's law?

vii. Show that the motion of simple pendulum is SHM.

viii. What do you mean by interference of light?

ix. Differentiate between year and light year.

x. Explain how cranes are able to lift very heavy load without toppling?

xi. Differentiate between elastic and inelastic collision by giving two examples of each.

xii. What are non-renewable sources of energy? Give three examples.

xiii. Explain the significance of moment of inertia in rotatory motion.

SECTION – C

Maximum Marks: 27

> Note: Note: Attempt any Three Questions. All questions carry equal marks:

Q.3 (a) Describe the diffraction of x-rays by crystal and derive an expression for Bragg's law.

(b) Find the polarizing angle for a glass of refractive index of 1.55.

Q.4 (a) What is heat engine? Describe its purpose, working and efficiency.

(b) The speed of sound in air at 0°C is 332 ms^{-1}. What will be the speed of sound at 22°C?

Q.5 (a) Explain torque. Derive an expression for torque acting on a particle due to a force F.

(b) A body of mass 10kg is dropped from a rest position 8m above the ground. What is its velocity at height of 6m above the ground?

Q.6 (a) Give three examples of illustrate the phenomenon of law of conservation of angular momentum.

(b) The moon revolves around the earth in almost a circle of radius 382400 km in 27.3 days. What is the centripetal acceleration?